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Do numerical probabilities promote informed stated preference responses under inherent outcome uncertainty? Insight from a coastal adaptation choice experiment

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ABSTRACT

Stated preference (SP) valuation provides insight into individuals' preferences for environmental goods and assigns a monetary value to these preferences. It employs survey questions that elicit information on choices or behaviors in response to hypothetical future scenarios involving changes to these goods. Outcome uncertainty (OU) within SP studies refers to uncertainty concerning whether the effects communicated in these hypothetical scenarios would actually occur, were the scenario to be implemented as indicated in the questionnaire. Inherent OU may be further defined as OU that is invariant across scenarios. For example, the effect of installing coastal flood defenses may depend on a probability of severe storms that is fixed in the study area over the relevant time horizon. Inherent OU has received little attention in the environmental SP literature. For example, it is unknown whether numerical, percentage probabilities—a ubiquitous means of communicating uncertainty in SP questionnaires—are an effective risk communication tool for inherent OU in environmental valuation scenarios. This article evaluates two treatments in a discrete choice experiment survey related to coastal climate change adaptation in Connecticut, USA: one provides only raw frequencies as a risk communication tool, while the other provides implied percentage probabilities in *addition* to the same raw frequencies. Results show that the use of percentage probabilities to communicate inherent OU has no additional effect on average welfare estimates or the choice behavior of respondents. These findings suggest that percentage probabilities may not provide useful information to SP respondents, beyond that conveyed by past event frequencies.

1. Introduction

Stated preference (SP) valuation provides insight into individuals' preferences via survey questions that elicit information on choices or behaviors that would occur under hypothetical scenarios, often involving changes to public or private goods relative to the status quo [1]. These methods are applied to estimate individuals' preferences and valuations (i.e., welfare measures, such as willingness to pay, WTP)—reflecting the economic value—of changes in environmental conditions that might arise. Discrete choice experiments (DCEs) are a common SP approach that asks survey respondents to make preference-based choices over a set of presented options—for example, alternative options for coastal flood adaptation. Each option (within each choice question) is characterized by a

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set of attributes, where one of the attributes typically describes direct monetary consequences to the household—for example, additional, binding taxes and fees that the household would be expected to pay to implement the proposed coastal adaptation option. Other attributes usually reflect non-monetary outcomes that are anticipated under each option—for example, measures of anticipated future flood impacts. Data on respondents' choices over options varying in environmental outcomes and costs allows preferences and welfare measures to be estimated [2–4].

Outcome uncertainty (OU) within SP studies refers to uncertainty concerning whether the outcomes communicated in these hypothetical future scenarios would actually occur, were the scenario to be implemented as indicated in the questionnaire. OU can be related to numerous factors, including uncertainty in scientific predictions or underlying uncertainty in ecological systems, among others.[1] Although the SP literature typically describes this situation as "outcome uncertainty," most cases can be simplified to a more restricted case of outcome risk, wherein a known probability distribution of possible outcomes is assumed. That is, researchers assign probabilities to possible outcomes within each option [6]. For example, respondents might be asked to choose between alternative management options for harmful algal blooms, each with a different probability that a bloom will occur over a specified time horizon (cf. [7]).

Most environmental SP surveys provide no formal communication or modeling of OU, often with the unstated assumptions that the presented scenario outcomes are certain, that the environmental outcomes in these scenarios reflect expected values, and/or that utilities depend only on the final states of these outcomes [7–10]. This stands in contrast to DCEs on health-related topics, for which uncertainty over outcomes (e.g., disease risk) is often a central focus [11]. Concerns that these assumptions can have important implications for the interpretation and validity of model results have sparked a growing body of environmental SP studies that incorporate OU in scenario designs and preference modeling.[2] Yet despite recent work, this literature has not yet reached consensus on best practices, including most effective means to communicate different types of OU within SP options [21,23,24].

Recent guidelines for SP studies [1] suggest that scenarios should communicate risk information in terms that are "readily understood" by respondents. There has been extensive prior research on risk communication and comprehension in general, including some prior work specific to SP studies [11,25–29]. Recommendations from this work vary, but often suggest the use of learning tools and graphic depictions to provide numerical (rather than verbal) risk information in a form that is widely accessible, for example, to respondents with varying numeracy skills and/or experience [30–35]. Despite this guidance, there have been few rigorous, large-sample tests of alternative risk communication approaches within environmental DCEs [36], with most DCE studies relying on simple percentage probabilities as the primary or sole risk-communication tool.[3] These observations have led to a general concern that the "presentation of risk in DCEs has not aligned with best practices in the risk communication literature" [11,26].

This article focuses on the question of whether numerical, percentage probabilities are an effective means to communicate OU within environmental DCEs. We focus specifically on the case of *inherent outcome uncertainty* [21], which may be defined as uncertainty that does *not vary within or across DCE scenarios*. This type of uncertainty is generally (though not necessarily) related to underlying or inherent uncertainty in ecological systems that cannot be altered through short-term human interventions, i.e., uncertainty that is "uncontrollable" over the relevant time horizon [23]. For example, the effect of coastal flood defenses may depend on the near-term probability of severe storms. This probability cannot (at least credibly) vary over DCE options that depict local coastal adaptation alternatives, because storm probability is not affected by this policy choice.

The implicit assumption underlying the ubiquitous use of percentage probabilities to communicate OU in DCEs—including inherent OU—is that respondents understand, interpret and use these probabilities as dictated by the standard neoclassical theory of choice, and that the resulting choices reveal their true underlying preferences for risky or uncertain outcomes. However, widespread evidence from economics and other disciplines suggests that individuals—influenced by factors such as numeracy skills, contextual cues, and prior experiences—may not necessarily interpret or use percentage probabilities as assumed by researchers.[4] Hence, there is a lack of clarity regarding whether reliance on percentage probabilities to communicate OU—the dominant approach in environmental DCEs—is warranted. It is also unclear whether percentage probabilities provide useful information to typical respondents, beyond that in other approaches to risk communication (e.g., based on relative event frequencies).

Questions of this type may be particularly important for inherent OU. The majority of environmental DCEs that account for *non-inherent* OU consider situations in which the probability of environmental outcomes varies over choice options presented to the re-

[1] An additional form of OU may arise due to low perceived consequentiality of survey responses, in which respondents perceive a less-than-certain probability that their survey responses will influence actual policies [5].

[2] See, for example, Zhai, Fukuzono, and Ikeda [12], Roberts, Boyer, and Lusk [7], Wielgus et al. [8], Rolfe and Windle [13,14], Ivanova, Rolfe, and Tucker [15], Glenk and Colombo [9,16], Wibbenmeyer et al. [17], Veronesi et al. [18], Reynaud and Nguyen [19], Torres, Bujosa, and Font [20], Torres, Faccioli, and Font [21], Meldrum et al. [22], and Aanesen et al. [23].

[3] For example, see Zhai, Fukuzono, and Ikeda [12], Roberts, Boyer, and Lusk [7], Rolfe and Windle [13,14], Ivanova, Rolfe, and Tucker [15], Glenk and Colombo [9,16], Wibbenmeyer et al. [17], Veronesi et al. [18], Reynaud and Nguyen [19], Torres, Faccioli, and Font [21], Glatt, Brouwer, and Logar [37], Meldrum et al. [22] and Davies, Wu and Schaafsma [24]. Still other studies use descriptive (non-numerical) risk indicators. This is done despite concerns regarding the subjective interpretation of these indicators (e.g., [33,35,38–41]). Lundhede et al. [10], for example, use qualitative risk descriptors (e.g., "very certain" and "rather certain") in a DCE generating implicit prices for bird conservation policies under climate change. Phillips [42] investigates beach visitors' preferences for coastal management options where the relative risk of flood damage to public and private property is a primary attribute that takes on the values "Low (1 in 20 years)", "Medium (1 in 10 years)," and "High (1 in 3 years)". Shaw and Baker [40] estimate Hurricane Katrina evacuees' WTP to avoid hurricane risks and use "low", "medium," and "high" to characterize the risk. Rolfe and Windle [43] value the benefits of controlling fire ants and embed verbal risk descriptors in choice option labels indicating "high" and "low" certainty of success.

[4] Examples of the literature in this area include Slovic [28,29]; Black, Nease, and Tosteson [44]; Yamagishi [45]; Windschitl and Weber [46]; Lipkus, Samsa, and Rimer [47]; Edwards, Elwyn, and Mulley [48]; Patt and Schrag [38]; Gilboa, Postlewaite, and Schmeidler [49]; Baker et al. [50]; Cameron, DeShazo, and Johnson [51]; and Sirota and Juanchich [32].

spondent. This is usually accomplished by embedding exogenously varying percentage probabilities in DCE options as standalone attributes or using probabilities as option framing variables [9,13,15,16,52]. In such cases, internal variation in the presented probabilities enables respondents to see clearly that some options and/or outcomes are associated with higher (or lower) probabilities of occurrence. Effects are typically evaluated using statistical tests that consider whether and how the presented probability information affects individuals' preferences and associated welfare measures, such as WTP. In other cases, split-sample studies compare the results of one DCE that includes exogenously varying percentage probabilities to those from an otherwise identical DCE that omits this information [7,24,53]. When OU is communicated, however, its importance to individuals' preferences is still implied directly via the direct inclusion of internal probability variation within each choice option.

Internal variation of this type is not present for inherent OU, which does not vary over choice options. There is also limited understanding of how respondents understand and process this type of scenario-invariant OU within environmental DCEs. Although a few studies have considered the effect of inherent OU within environmental SP studies in general [6,8,21,23,54], we are aware of no research that evaluates the effect of alternative risk communication formats within studies of this type. This reflects a more general limitation within the DCE literature on OU (inherent or otherwise): nearly all of these studies consider only one possible form of risk communication, without considering whether and how results might differ as a function of alternative approaches.

To address this general lack of insight in the literature, this study implements a split-sample DCE to compare response properties and welfare estimates under alternative approaches of communicating the same inherent OU, for otherwise identical policy options. Because we consider inherent OU, relevant probabilities do not vary over the presented choice options. The goal is a formal test of whether the use of percentage probabilities to communicate the same inherent OU (compared to alternative communication approaches) leads to different welfare estimates, response properties, or symptoms of scenario rejection/complexity, such as increased choice randomness. That is, we specify the same level of (fixed) inherent OU across survey treatments, but use different approaches to communicate this uncertainty across the treatments.[5]

The study is implemented using an application of DCEs to coastal flood adaptation in Connecticut, USA, where adaptation outcomes depend on the (inherently) uncertain future occurrence of severe coastal storms. Data are gathered using two otherwise identical survey treatments. Both survey treatments present information on historical storm frequency in the study area, but one also presents the percentage probabilities consistent with this historical frequency—following the common approach of presenting probabilities in percentage form. Comparison of results across these treatments provides insight into the effect of percentage probabilities as a tool for communicating inherent OU—and tests explicitly whether and how this additional risk communication form influences DCE results. Two models on pooled data from the two survey treatments are estimated: a mixed logit model in WTP space and a latent class model in WTP space. Each model allows for systematic variation in WTP estimates associated with the survey treatments to study the potential effect of percentage probabilities on individuals' choices and WTP values.

Results suggest that the addition of information on percentage probabilities to the questionnaire has no effect on welfare estimates, compared to those derived from the questionnaire that presented only raw storm frequencies. These results suggest that in at least some types of environmental SP questionnaires, the use of percentage probabilities to convey information on inherent OU may not provide useful information to respondents, beyond the information conveyed via past event frequencies. These findings highlight the need for further research on OU communication in valuation surveys.

2. Risk communication and inherent outcome uncertainty in discrete choice experiments

As described by Torres et al. [21], inherent OU is "the component of environmental uncertainty which derives from the stochastic nature of an ecosystem's behavior as a result of interactions between physical, chemical, ecological and human factors [55,56]". From the perspective of SP welfare elicitation, this type of uncertainty (or risk) is not typically considered to be controllable over the relevant time horizon via management or policy interventions, and hence cannot vary across DCE scenarios (or at least across those scenarios that reflect realistic and credible possibilities). Uncertainties associated with many environmental phenomena—such as those associated with climate change impacts—are understood to be an irreducible aspect of socio-ecological systems [57,58]. Because these uncertainties cannot be significantly reduced by scientific research or influenced by management (at least during the time horizon covered by most valuation scenarios), they cannot be credibly presented as variable within SP questionnaires [21].

Examples of inherent OU are common within environmental settings but are rarely made explicit within SP valuation. For instance, the effect of management options to sustain coastal marshes depends on uncertain sea-level rise, and this inherent OU does not vary as a function of local management actions [59]. The outcomes of coastal flood adaptation (such as building sea walls) may depend on the likelihood of future coastal storms, where this underlying probability is not affected by local adaptation actions [6]. Torres et al. [21] consider yet another case, in which management effects on bird species depend on irreducible, ecological inherent OU.

Only a few SP studies consider cases of inherent OU [6,8,21,54]. Even fewer [6] consider cases of true inherent OU, in which the uncertainty does not vary either over choice options or different survey versions (treatments). In cases such as these, the presentation of probabilities (describing inherent OU) that vary across either survey versions or options may strain the credibility of survey scenarios, leading to scenario rejection caused by presented probabilities that respondents deem to be irrelevant, unrealistic, or inconsistent with their priors [51,60–63].[6]

[5] To avoid confounding effects associated with embedding percentage probabilities as an *additional* attribute in choice scenarios (e.g., whether observed differences in welfare estimates are due to increased choice task complexity or the use of percentage probabilities *per se*), the questionnaire presents this information on a separate survey page preceding the DCE tasks. The tasks themselves are identical—with identical complexity—across DCE treatments.

[6] For example, presenting respondents with counterfactual, unrealistic information on the near-term probability of severe storms in their local area might cause them to reject the information presented in the questionnaire.

As noted above, environmental DCEs that consider inherent OU typically communicate it using percentage probabilities. In some cases, this is supported by visual aids to help respondents understand the presented information [21]. External tests (when inherent OU varies across different survey treatments) typically find that the inclusion of—and variation in—percentage probabilities influences DCE results in predictable ways [8,21,54]. This follows similar findings for more general types of OU in SP valuation. However, findings such as these do not necessarily imply that DCE results (e.g., WTP estimates) will be robust to alternative means of communicating the same inherent OU.

Findings from the broader literature on risk and uncertainty further suggest that risk perceptions and associated behaviors can be sensitive to both context and communication approach. These results suggest that respondents may not always interpret OU information in the ways often assumed by the DCE literature. Past studies show, for example, that a 30 % chance of snow in December may convey a higher degree of certainty than a 30 % chance of snow in November [32,46]. Yamagishi [45] shows that participants (counterintuitively) rated cancer as riskier when it was described as killing 1286 out of 10,000 than as killing 24.14 out of 100 people. Harris and Corner [64] report that "probability expressions referring to severe events are interpreted as denoting higher probabilities than those referring to more neutral events". The tendency of individuals to interpret or use percentage probabilities in ways that are unanticipated by researchers may be increased if the presented probabilities conflict with individuals' prior perceptions or motivations, leading to adjustment or rejection of the provided information [51,65].[7]

Despite these possibilities, and to the best of the authors' knowledge, no prior SP study dealing with inherent OU compares responses or welfare estimates across survey versions whose sole difference is the approach used to communicate otherwise identical inherent OU. This lack of insight is not trivial. In the absence of external (e.g., split-sample) tests that evaluate the effect of communicating inherent OU using percentage probabilities, it is difficult to draw credible conclusions on whether these are an effective form of risk communication within SP studies.

3. The discrete choice experiment survey

3.1. Survey development

To empirically test the effect of communicating inherent OU in DCEs using percentage probabilities, data was generated using a DCE survey designed to probe into individuals' preferences for coastal climate change adaptation policies in Old Saybrook, Connecticut, USA. The survey instrument was developed over two years in a collaborative process involving economists and natural scientists; meetings with town planners, engineers and stakeholder groups; and focus groups with community residents. Attention to development and testing helped ensure that survey language, graphics, and format were understood by respondents; that respondents and researchers shared interpretations of survey terminology and scenarios; and that choice questions and their attributes and levels (described below) captured adaptation outcomes viewed as relevant and realistic.[8]

The DCE part of the survey hosts three choice questions. A sample choice question is shown in Fig. 1. For every choice question, each household thus chooses among three choice options (or adaptation options): Two multi-attribute adaptation options (*A, B*) at some cost to the household, and a status quo option that involves no new adaptation actions and zero household cost.

As Fig. 1 shows, each adaptation option within each choice question is characterized by attributes that reflect outcomes expected by the mid-2020s. These outcomes are described in terms of attributes characterizing (a) the share of homes vulnerable to flooding in a Category 2 storm; (b) the additional share of homes vulnerable to flooding *only* in a Category 3 or greater storm (henceforth referred to simply as 'Category 3 storm'); (c) wetland acreage expected to be lost because of flooding or erosion; (d) natural beach and dune acreage expected to be lost because of flooding or erosion; (e) the share of the coastline length to be hard-armored with sea walls; and (f) the emphasis of the adaptation plan on hard structures or soft (natural) measures compared to current approaches. Finally, the DCE includes an attribute indicating annual household cost in unavoidable taxes and fees.

Baselines and policy levels for these attributes were grounded in inundation and adaptation scenarios identified by digital inundation models (i.e., future storm and flooding scenarios) developed for the Coastal Resilience decision-support platform (www.coastalresilience.org), combined with expert consultations. Following the guidelines of Johnston et al. [67] and Schultz et al. [68] for biophysical indicators within DCEs, all non-cost continuous attributes presented each adaptation method and outcome in relative (percentage) terms with regard to upper and lower reference conditions (i.e., best and worst possible in Old Saybrook), and were defined in survey informational materials. Relative attribute levels represented losses approaching the upper reference condition (100 % loss), starting from the lower reference condition (0 % loss). Scenarios also displayed the cardinal basis for relative levels where applicable [69]. For example, the attribute representing the number of homes expected to flood in a Category 2 storm was presented both as a cardinal number (e.g., 1411 of the current 5034 homes) and as a percentage relative to the total number of homes in Old Saybrook (e.g., 28 %). Descriptive statistics for attribute levels are provided below in Section 3.3.

Prior to presenting choice questions, the survey provided information describing tradeoffs associated with alternative approaches to coastal adaptation, projected inundation scenarios in the mid-2020s, and baseline (status quo) effects with no new adaptation ac-

[7] Multiple works within the SP literature consider the ramifications of presenting respondents with risk information that conflicts with their subjective understanding, including biased estimates and scenario adjustment/rejection (e.g., [51,66]; Howard et al., 2020). Prior work also suggests that individuals' motivations may influence reactions to probability information, with individuals favoring formats (such as verbal presentations) that may be redefined (or adjusted) to meet predetermined motivations [65].

[8] Adaptation outcomes shown in the DCE questions were informed using data provided by the Center for Climate Systems Research at Columbia University, NASA's Goddard Institute for Space Studies, the National Oceanic and Atmospheric Administration (NOAA) Services Center, and The Nature Conservancy. Attributes were selected based on a conceptual model combining input from focus groups; natural scientists with expertise in sea-level rise, coastal vulnerability and adaptation; coastal flooding scenarios; and interviews with community officials.

PROTECTION OPTION A and **PROTECTION OPTION B** are possible protection options for Old Saybrook. **NO NEW ACTION** shows what is expected to occur with no additional protection. All plans involve hard and soft defenses in different areas. Given a choice between the three, how would you vote?

Methods and Effects of Protection	Result in 2020s with NO NEW ACTION	Result in 2020s with PROTECTION OPTION A	Result in 2020s with PROTECTION OPTION B
	No Change in Existing Defenses	More Emphasis on SOFT Defenses	More Emphasis on SOFT Defenses
Homes Flooded in Category 2 Storm	28% 1,411 of 5,034 homes expected to flood in a Category 2 storm	32% 1,611 of 5,034 homes expected to flood in a Category 2 storm	28% 1,411 of 5,034 homes expected to flood in a Category 2 storm
Homes Flooded Only in Category 3+ Storm	23% 1,174 of 5,034 homes expected to flood only in a Category 3+ storm	23% 1,174 of 5,034 homes expected to flood only in a Category 3+ storm	27% 1,359 of 5,034 homes expected to flood only in a Category 3+ storm
Wetlands Lost	5% 25 of 497 wetland acres expected to be lost	2% 10 of 497 wetland acres expected to be lost	2% 10 of 497 wetland acres expected to be lost
Beaches and Dunes Lost	10% 3 of 30 beach acres expected to be lost	16% 5 of 30 beach acres expected to be lost	16% 5 of 30 beach acres expected to be lost
Seawalls and Coastal Armoring	24% 12 of 50 miles of coast armored	24% 12 of 50 miles of coast armored	24% 12 of 50 miles of coast armored
$ Cost to Your Household per Year	$0 Increase in annual taxes or fees	$125 Increase in annual taxes or fees	$95 Increase in annual taxes or fees
HOW WOULD YOU VOTE? (CHOOSE ONLY ONE) I vote for	☐ I vote for NO NEW ACTION	☐ I vote for PROTECTION OPTION A	☐ I vote for PROTECTION OPTION B

Fig. 1. Sample DCE question.

tions. This introduction, presented over the initial pages of an 18-page survey booklet, used a combination of text, graphics including geographic information system (GIS) maps, and photographs to convey essential background information. Detailed instructions were also provided, including reminders to consider budget constraints and statements highlighting the survey's consequentiality [70]. After describing the attributes and showing an example of a DCE question, each of the three DCE questions to be answered were presented separately on consecutive pages. The survey concluded with attitudinal and sociodemographic questions. Survey language and graphics were subject to extensive pretesting in 13 focus groups and multiple cognitive interviews [71,72], including the use of verbal protocols to gain insight into respondents' comprehension and decision processes [73]. The study was reviewed and approved by the Clark University Institutional Review Board (IRB), protocol #2013-071.

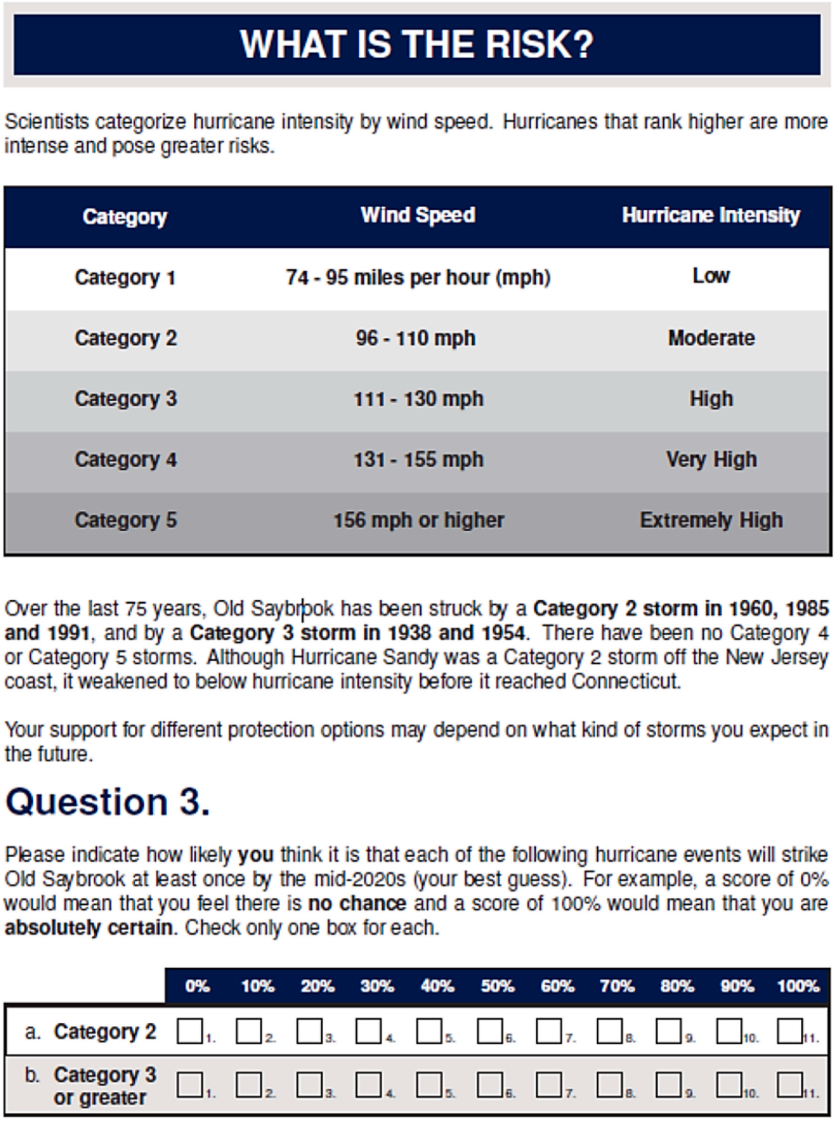
3.2. Split-sample survey treatment: communicating risk information

To test whether and how percentage probabilities affect welfare estimates, two treatments (or versions) of the survey were developed. The first treatment, S1, provided only the historical frequencies of Category 2 and 3 storms (based on the Saffir-Simpson Hurricane Wind Scale)[9] over the past 75 years, but did not present corresponding percentage probabilities. These frequencies were presented to respondents as shown in Fig. 2.

The second survey treatment, S2, provided the same (identical) information on historical storm frequencies but also supplemented this information with corresponding percentage probabilities for each of these two storm events, rounded to the nearest 5 %.[10] These were calculated from long-term storm frequency data that implied a 55 % and 20 % chance that a Category 2 and Category 3 storm,

[9] This scale distinguishes five categories of hurricane intensity, from Category 1 (least intense, with sustained winds of 74–95 mph) to Category 5 (most intense, with sustained winds of 157 mph or higher). From National Oceanic and Atmospheric Administration, http://www.nhc.noaa.gov/aboutsshws.php, accessed May 13, 2021.

[10] The study design did not include a group for which no information on storm probability was provided (i.e., a "no probability information" control group). This treatment was excluded for two reasons. First, respondents in pretest focus groups requested this information when it was omitted, suggesting that some information of this type was necessary to ensure valid, well-informed responses (and to prevent respondent frustration with a perceived lack of salient information in the questionnaire). Second, multiple split-sample studies in the literature already demonstrate that the inclusion of (scenario-invariant) information on OU influences responses, compared to otherwise identical questionnaires where information on OU is excluded [21,23].



WHAT IS THE RISK?

Scientists categorize hurricane intensity by wind speed. Hurricanes that rank higher are more intense and pose greater risks.

Category	Wind Speed	Hurricane Intensity
Category 1	74 - 95 miles per hour (mph)	Low
Category 2	96 - 110 mph	Moderate
Category 3	111 - 130 mph	High
Category 4	131 - 155 mph	Very High
Category 5	156 mph or higher	Extremely High

Over the last 75 years, Old Saybrook has been struck by a **Category 2 storm in 1960, 1985 and 1991**, and by a **Category 3 storm in 1938 and 1954**. There have been no Category 4 or Category 5 storms. Although Hurricane Sandy was a Category 2 storm off the New Jersey coast, it weakened to below hurricane intensity before it reached Connecticut.

Your support for different protection options may depend on what kind of storms you expect in the future.

Question 3.

Please indicate how likely **you** think it is that each of the following hurricane events will strike Old Saybrook at least once by the mid-2020s (your best guess). For example, a score of 0% would mean that you feel there is **no chance** and a score of 100% would mean that you are **absolutely certain**. Check only one box for each.

	0%	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
a. Category 2	☐ 1.	☐ 2.	☐ 3.	☐ 4.	☐ 5.	☐ 6.	☐ 7.	☐ 8.	☐ 9.	☐ 10.	☐ 11.
b. Category 3 or greater	☐ 1.	☐ 2.	☐ 3.	☐ 4.	☐ 5.	☐ 6.	☐ 7.	☐ 8.	☐ 9.	☐ 10.	☐ 11.

Fig. 2. S1 survey treatment: Historical storm frequency.

respectively, would strike Old Saybrook at least once by the year 2025. These percentage probabilities were presented to respondents as shown in Fig. 3.

Frequencies and percentage probabilities were both presented in a dedicated section of the survey (preceding the DCE questions) that provided information on storm uncertainty and identified areas of the community that would be expected to flood under storms of different intensities, under the status quo (i.e., with no new flood adaptation measures in the affected area). Although the presented storm frequencies (in both S1 and S2) and percentage probabilities (only in S2) were accurate and factually consistent, the two types of information are not trivially identical. This was purposeful and reflected the fact that percentage probabilities (typically derived from long-term probability distributions) and short-term observed frequencies provide different perspectives on the same underlying uncertainty. Hence, it was not possible for respondents to directly calculate the presented storm probabilities (in treatment S2) using *only* the presented 75-year storm frequency information. Additional information was required. As of 2014, the true long-period average annual probability of a Category 3 storm striking Connecticut was estimated to be 2 % [74]. This corresponded to a 20 % chance of at least one Category 3 storm during the 11 years from 2014 to 2025, which was the percentage presented in treatment S2.[11] To reiterate, since one cannot calculate this long-period average using data from only the 75-year period anchored on a known storm date

[11] This may be calculated as one minus the probability that no storm will occur in the region over 11 successive years. For a Category 3 storm this is $1 - 0.98^{11} = 0.20$. The remaining 20 % probability reflects the chance that one or more storms will occur during this time.

WHAT IS THE RISK?

Scientists categorize hurricane intensity by wind speed. Hurricanes that rank higher are more intense and pose greater risks.

Category	Wind Speed	Hurricane Intensity
Category 1	74 - 95 miles per hour (mph)	Low
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Category 4	131 - 155 mph	Very High
Category 5	156 mph or higher	Extremely High

Over the last 75 years, Old Saybrook has been struck by a **Category 2 storm in 1960, 1985 and 1991**, and by a **Category 3 storm in 1938 and 1954**. There have been no Category 4 or Category 5 storms. Although Hurricane Sandy was a Category 2 storm off the New Jersey coast, it weakened to below hurricane intensity before it reached Connecticut.

Based on past storm events, scientists estimate that there is approximately a **55% (or about one in two) chance that a Category 2 storm will strike Old Saybrook at least once by the mid-2020s** (0% would mean there is no chance and 100% would mean it is absolutely certain).

In contrast, scientists estimate that there is approximately a **20% (or one in five) chance that a Category 3 or higher storm will strike Old Saybrook at least once by the mid-2020s**.

Fig. 3. S2 survey treatment: Historical storm frequency and corresponding percentage probabilities.

(1938),[12] although treatments S1 and S2 convey information on the same inherent OU drawn from historical weather patterns, they do so from different perspectives. Because both treatments present underlying information on past storm frequencies, comparison of results across these two treatments enables a well-identified test on whether the provision of the supplemental percentage probabilities influenced responses. Other than this differences in risk communication – presented on a single page of the survey (see Figs. 2 and 3) – the two DCE surveys were identical, including the attributes in the choice questions and their relative levels.

Percentage probabilities are linked to the choice questions only through the number of homes that would be flooded conditional on a Category 2 storm and the number of *additional* homes that would be flooded conditional on a Category 3 storm, both forecast as of the year 2025, where these storms have different inherent probabilities of occurrence. Other choice attributes, such as acres of wetlands or beach expected to be lost (as defined in Table 1 below) are more heavily influenced by ongoing erosion and sea-level rise rather than by instances of acute flooding, and thus the changes shown would occur regardless of storm occurrence. These differences were made clear in the questionnaire. To ensure that respondents understood relationships between storms of different intensities and home flooding, the survey included a simple graphic illustrating the risk-related differences between these two groups of homes. This figure also clarified the definition of these two groups of homes, highlighting, for example, that they were independent and non-overlapping groups.

3.3. Attributes and levels: descriptive statistics and experimental design

Table 1 provides a definition of the attributes that described choice options in the DCE and the attribute levels used in the survey. The table also defines the variables that capture the attribute levels and are subsequently used in the data modeling, and descriptive statistics for the variables in the final data for both the S1 and S2 survey versions. Values of the statistics, including means and standard deviations, are determined by the DCE experimental design and the final set of questionnaires completed by respondents (see details below). For example, over all returned surveys for treatment S1, the mean of attribute *homes2* in the final data is 26.18.

Grounded in these attribute levels, a fractional factorial experimental design was generated using a D-efficiency criterion [75–78] for main effects and selected two-way interactions, yielding 72 profiles blocked in 24 booklets. S-efficiency was also used to evaluate

[12] This 75-year period was chosen because it spanned the difference between the well-known hurricane of 1938 in New England and the year of survey implementation (2014), rounded down to the nearest 5 years. Storm frequencies have been often discussed in New England since 1938 due to the major hurricane that occurred that year.

Table 1
DCE Attributes, their Levels, and Descriptive Statistics (means and standard deviations include the status quo option of no adaptation action).

Choice Attribute and Model Variable Name	Definition	Attribute/ Variable Levels	S1 Survey (Omitted Numerical Probabilities) Mean (Std. Dev.)	S2 Survey (Included Numerical Probabilities) Mean (Std. Dev.)
status quo	Alternative specific constant associated with the status quo (i.e., a choice of neither Option A or Option B).	0; 1	0.33 (0.47)	0.33 (0.47)
homes2	The percentage of Old Saybrook homes that are expected to flood in a Category 2 or higher storm in the mid-2020s. With no new action, 28 % of homes (1411 of the current 5034 homes in Old Saybrook) will be in this higher risk category by the mid-2020s.	20 %; 24 %; 28 %; 32 %	26.18 (3.90)	26.42 (3.88)
homes3	The percentage of Old Saybrook homes that are expected to flood ONLY in a Category 3 or higher storm in the mid-2020s. They are not expected to flood in a Category 2 storm. With no new action, 23 % of homes (1174 of the current 5034 homes in Old Saybrook) will be in this moderate risk category by the mid-2020s.	16 %; 19 %; 23 %; 27 %	21.97 (3.46)	21.77 (3.42)
wetlands	The percentage of Old Saybrook's coastal marshes expected to be lost by the mid-2020s due to flooding or erosion. With no new action, 5 % of Old Saybrook's coastal marshes (25 of 497 acres that exist today) are expected to be lost.	2 %; 5 %; 10 %	5.05 (2.63)	5.06 (2.61)
beaches	The percentage of Old Saybrook's beaches and dunes expected to be lost by the mid-2020s due to flooding or erosion. With no new action, 10 % of Old Saybrook's beaches and dunes (about 3 of 30 acres that exist today) are expected to be lost.	3 %; 5 %; 7 %; 10 %	9.53 (4.28)	9.40 (4.32)
seawalls	The percentage of Old Saybrook's coast shielded by hard defenses. With no new action, 24 % of Old Saybrook's coastline (12 of 50 miles) will have hard defenses by the mid-2020s. This is the same level as today.	15 %; 24 %; 35 %	24.94 (6.29)	24.77 (6.39)
hard	Binary variable that takes on a value of 1 if the adaptation option emphasizes hard defenses or shoreline armoring and a value of 0 otherwise.	0; 1	0.19 (0.39)	0.20 (0.40)
soft	Binary variable that takes on a value of 1 if the adaptation option emphasizes soft or natural defenses and a value of 0 otherwise.	0; 1	0.16 (0.37)	0.16 (0.37)
cost	How much the option will cost the household per year in unavoidable taxes and fees. These funds were assumed to be legally guaranteed to be spent only on the coastal protection option that the household voted for.	$0; $35; $65; $95; $125; $ 155	62.46 (56.49)	62.45 (56.14)

minimum sample sizes required to estimate preference parameters [77,78]. Each respondent was provided with three choice questions and was instructed to consider each choice question as independent and non-additive.

3.4. Survey implementation

The mail-back survey questionnaires were sent to 576 randomly selected Old Saybrook households via U.S. mail during May through July 2014 together with a postage-paid return envelope, with repeated mailings following Dillman et al. [79] to increase response rates. Names and addresses for survey invitation mailings were obtained using random draws from an address-based residential occupancy list derived from the US Postal Service Delivery Sequence File, obtained from a professional vendor. No stratification or additional quotas were applied to the sample. The mailing sample size was determined by anticipated sample sizes required to estimate model parameters (based on S-efficiency estimates for the experimental design; see above) and the anticipated response rate, along with the budget available for survey implementation. The analysis is based on 146 returns from the 493 deliverable S1 surveys, for a response rate of 29.6 %, and 123 returns from the 508 deliverable S2 surveys, for a 24.2 % response rate.

To reiterate, the two treatments of the survey differ only in the communication of inherent OU. The first treatment, S1, provided only the historical frequencies of Category 2 and 3 storm events over the preceding 75 years (Fig. 2). The second survey treatment, S2, supplemented this information with percentage probabilities for each of these two storm events (by 2025), calculated based on long-term historical frequencies (Fig. 3). These probabilities are linked to the choice scenarios through the attributes *homes2* and *homes3* (see Table 1), which indicate the share of homes expected to flood in a Category 2 storm, and the share of additional homes expected to flood in a Category 3 storm, respectively. The key research objective is to test whether the provision of percentage probabilities lead to different choice behaviors and different WTP values for SP adaptation outcomes, particularly the protection of homes vulnerable to Category 2 and Category 3 storms.

4. Econometric models

Specification of the econometric models follows a traditional random utility framework [80,81], in which household h chooses their preferred option among three coastal adaptation policy options ($p = \{A, B, N\}$), with two options involving policy changes (A and B) at some cost to the household and a third status quo option (N) assuming no adaptation action and zero household cost. Every household faces three independent choice questions or scenarios ($j = \{1, 2, 3\}$). Within each scenario, each choice (adaptation) op-

tion is described by a vector of variables $\boldsymbol{X}_{pjh}$ representing non-monetary characteristics (attributes) of the adaptation policy and variable C_{pjh} expressing the unavoidable household cost of the policy. The variables are included in the models in the forms defined in Table 1.

We apply two standard approaches to account for possible heterogeneity in households' preferences and WTP: a mixed (random parameters) logit model [82,83] and a latent class model [84]. While the former assumes that preference heterogeneity can be characterized by continuous distributions of preference or WTP parameters, the latter considers discrete distributions of the parameters.[13] The literature is inconclusive about which modeling approach is superior, and in practice the choice between these two approaches depends on the analyst's assumptions regarding the form of preference heterogeneity likely to be most prevalent within the data. We estimate both models in WTP space, allowing parameter estimates to be interpreted as direct expressions of dollar-denominated marginal values for changes in respective attributes.

To address our primary research question, these models are estimated using pooled data from both DCE treatments (S1 and S2). Each model includes a dummy variable S_h that identifies possible effects related to the survey treatment received by a household (that is, whether the survey questionnaire provided percentage probabilities to communicate inherent OU, as in the S2 treatment, or whether it did not, as in the S1 treatment). Given a lack of *a priori* knowledge on how the form of risk information communication might influence choice behavior, we allow for two alternative ways that this might occur. In the mixed logit model, the survey treatment variable is used to examine potential shifts in the mean parameter estimates. In the latent class model, the treatment variable enters the class membership probability function together with four sociodemographic variables. The subsections below discuss and contrast the two approaches.

4.1. Mixed (random parameters) logit model

To develop the mixed logit WTP-space specification, we represent the utility of household h, from policy scenario p, in choice task j as

$$U_{pjh}(\bullet) = V_{pjh}\left(\boldsymbol{X}_{pjh}, C_{pjh}\right) + e_{pjh} = \boldsymbol{\beta}_h{}'\boldsymbol{X}_{pjh} - \alpha_h C_{pjh} + e_{pjh}, \tag{1}$$

where $V_{pjh}(\bullet)$ is a function representing an observable component of the utility and e_{pjh} is an unobservable, or stochastic, component of the utility typically modeled as an econometric error. When choosing between options *A, B,* and *N*, with the utility specified as in (1), the household is assumed to choose the option which offers the greatest expected utility. As shown in the right-hand side of (1), such models are typically defined using an additively separable, linear functional form for utility, where α_h is the marginal utility of income (derived as a negative of the marginal utility of cost attribute C_{pjh}) and $\boldsymbol{\beta}_h$ is a conforming vector of marginal utilities associated with non-monetary attributes $\boldsymbol{X}_{pjh}$. All marginal utility parameters are household-specific, as suggested by indexing over h, which allows for heterogeneous preferences among households—a characteristic distinguishing the mixed logit approach.[14]

The underlying model in (1) may be estimated in either preference space or WTP space [85]. Mixed logit models in WTP space circumvent challenges associated with deriving welfare measures (such as WTP) from their preference- or utility-space counterparts [86–88],[15] while obtaining welfare estimates is of major interest in most valuation studies. Thus, the analysis is based on a mixed logit model in WTP space. To derive the WTP-space model, we first divide all arguments in (1) by the logit scale parameter μ_h, to obtain

$$U_{pjh}(\bullet) = \boldsymbol{\gamma}_h{}'\boldsymbol{X}_{pjh} - \lambda_h C_{pjh} + \varepsilon_{pjh}, \tag{2}$$

where $\boldsymbol{\gamma}_h = \beta_h/\mu_h$ is a vector of preference-space coefficients on non-monetary policy attributes $\boldsymbol{X}_{ph}$, $-\lambda_h = -\alpha_h/\mu_h$ is a preference-space coefficient on the policy cost, and the resulting error term ε_{pjh} has an i.i.d. type I extreme value distribution with constant variance $var\left(\varepsilon_{pjh}\right) = \pi^2/6$ [85,88].

The vector of marginal WTP estimates (or implicit prices) for a change in non-monetary attributes may be calculated as a ratio of the coefficients on these non-monetary attributes and the cost coefficient, $\boldsymbol{\omega}_h = \gamma_h/\lambda_h = \beta_h/\alpha_h$. We hence rewrite (2) to obtain the WTP-space specification [85],

$$U_{pjh}(\bullet) = \lambda_h\left[\left(\boldsymbol{\gamma}_h/\lambda_h\right)'\boldsymbol{X}_{pjh} - C_{pjh}\right] + \varepsilon_{pjh} = \lambda_h\left(\boldsymbol{\omega}_h'\boldsymbol{X}_{pjh} - C_{pjh}\right) + \varepsilon_{pjh}, \tag{3}$$

which is behaviorally equivalent to the preference-space specification in (2). The elements of vector $\boldsymbol{\omega}_h$ are random coefficients representing estimates of marginal WTP, assumed to be normally distributed. To ensure a positive marginal utility of income, we specify $\lambda_h = e^{\nu_h}$, where ν_h is the underlying latent normal factor that defines the lognormally distributed cost coefficient [86,88].[16]

Based on this underlying model, we extend (3) to allow for systematic variation in $\boldsymbol{\omega}_h$ between the S1 and S2 treatments. Recall that for treatment S2, respondents are provided with the percentage probability of storms, in addition to information on historical

[13] In theory, a mixed logit model can accommodate any discrete or continuous distribution of model parameters. However, continuous parameter distributions (e.g., normal, lognormal) are most commonly used.

[14] Assuming, instead, that parameters do not differ across households implies homogenous preferences and leads to a multinomial logit specification.

[15] The challenges are related particularly to the randomly specified cost coefficient [83,85,86,89,90]; (cf. [91]), whose presence in the denominator of the analytical WTP expression may lead to behaviorally and statistically implausible WTP values, for instance, due to infinite WTP moments [89]. Cost coefficient distributions that lead to finite WTP moments can lead to additional analytical challenges, such as problems associated with the long tails of the lognormal distribution typically assumed for the cost coefficient [83].

[16] In WTP-space models, the random cost parameter, denoted here as λ_h, captures variation in both marginal utility of income and scale [86].

storm frequencies. For treatment S1, only historical frequencies are presented. To allow for the possible systematic variation in parameter estimates between the S1 and S2 samples—helping identify the effect of percentage probabilities as a means of communicating inherent storm risk—we specify the vector of WTP parameters in (3) as

$$\boldsymbol{\omega}_h = \boldsymbol{\omega}_h^* + \boldsymbol{\rho} S_h, \tag{4}$$

where $\boldsymbol{\omega}_h^*$ has a multivariate normal distribution with a set of means and a covariance matrix to be estimated; $\boldsymbol{\rho}$ is a vector of parameters to be estimated; and S_h is an indicator variable that takes a value of zero if household h received the S1 survey (which omits percentage probabilities) and a value of one if household h received the S2 survey (which provides percentage probabilities).

Using the same notation, we redefine the cost coefficient as

$$\lambda_h = e^{(v_h + \tau S_h)}, \tag{5}$$

with parameter τ to be estimated. Equation (5) enables marginal utility of income and scale to vary systematically depending on the survey treatment.

Within this model, the key research question relates to the vector of parameters $\boldsymbol{\rho}$, reflecting the effects of different communication formats of inherent OU. Specifically, we focus on whether the provision of this information as a percentage probability ($S_h = 1$) leads to variations in $\boldsymbol{\omega}_h$ (WTP estimates). The presence of a significant effect would suggest that the provision of percentage probabilities (in addition to storm frequencies) led to different welfare estimates, at least potentially reflecting better-informed choices by respondents.

4.2. Latent class multinomial logit model

To develop the latent class multinomial logit model in WTP space, the utility of household h from choosing policy scenario p in choice task j can be expressed in a similar way to the earlier representation for the mixed logit model in (3), that is, as

$$U_{pjh}^c(\bullet) = \lambda_c \left(\boldsymbol{\omega}_c' \boldsymbol{X}_{pjh} - C_{pjh} \right) + \varepsilon_{pjh}. \tag{6}$$

The notation in (6) is analogous to that introduced in the earlier subsection on the mixed logit model, with the only difference being the indexing of utility by class c ($c = \{1, 2, \ldots, C\}$) to which a household belongs. In the latent class model, WTP parameters and the cost coefficient are class-specific. Naturally, and as in reality, we are not able to determine with certainty to which class a household belongs, but we can estimate the probability with which a household is a member of a given class.

As introduced earlier, we define the probability of belonging to class c as a function of the survey treatment received by a household and sociodemographic characteristics. Consequently, the class membership probability is explained by variable S_h, specified as above, and sociodemographic characteristics in vector $\boldsymbol{SD}_h$. Thus, the probability of belonging to class c, π_c, can be represented

$$\pi_c = \frac{\exp\left(\delta_c + \kappa_c S_h + \boldsymbol{\varphi}_c' \boldsymbol{SD}_h\right)}{\sum_{m=1}^{C} \exp\left(\delta_m + \kappa_m S_h + \boldsymbol{\varphi}_m' \boldsymbol{SD}_h\right)} \tag{7}$$

where δ_c, κ_c and $\boldsymbol{\varphi}_c$ are parameters to be estimated for $c = \{1, 2, \ldots, C-1\}$. For identification, the parameters for the last class (i.e., class C) are equal to zero, that is, $\delta_C = 0$, $\kappa_C = 0$ and $\boldsymbol{\varphi}_c = 0$. The parameters are therefore interpreted relative to class C, which constitutes a reference (or default) category.

Within the latent class model, the key research question relates to the parameter κ_c which allows WTP values to vary via the effect of S_h on the latent class membership. Here, we focus on whether the provision of percentage probabilities ($S_h = 1$) leads to differences in the probability of falling into different classes—and what this implies about choice behavior and welfare estimates.

4.3. Model estimation

The mixed logit model is estimated using the simulated maximum likelihood method with 6000 Sobol draws, with mean WTP and cost coefficients specified as correlated and random with normal distributions for WTP parameters and a lognormal distribution for the cost parameter. The latent class model is estimated via a maximum likelihood method. To ensure model convergence, all continuous attribute variables are scaled prior to estimation by dividing non-monetary attribute levels by 10 and policy cost by 100.[17] Several other specifications were estimated to ensure robustness of results, all of which have generated results consistent with those shown in this article. The selection of models presented in the article is guided by model fit based on the value of the log-likelihood function and Akaike information criterion (AIC). The econometric analysis is conducted in Matlab.[18]

[17] This is a common approach for models of this type and has no substantive impact on the model results other than to promote convergence.

[18] We use a custom code developed in Matlab, which is available at https://github.com/czaj/DCE under CC BY 4.0 license.

5. Results

This section presents the results of the mixed logit and latent class models used to answer research questions regarding the potential effect of a risk communication format on choice behavior and welfare estimates.[19] As described above, these tests are underpinned by an assumption that is seemingly implicit in the environmental SP literature—that the presentation of risk information in the form of percentage probabilities will lead to more informed responses and welfare estimates, or at least to different responses than would be observed in the absence of this information. It is not straightforward to determine *a priori* which econometric approach—the mixed logit model or the latent class model—matches the data better. We argue that each may provide distinct insights into the effect of the survey treatments in question. Hence, we proceed with a discussion and comparison of results from both models.

5.1. Mixed (random parameters) logit model

Results of the mixed logit model are shown in Table 2. Mean WTP coefficient estimates have anticipated signs and are consistent with preferences revealed during focus groups. Because non-monetary continuous attributes (that is, *homes2*, *homes3*, *wetlands*, *beaches* and *seawalls*) are specified as resource losses, negative WTP values are expected. We thus conclude that the signs are in line with standard neoclassical assumptions. However, not all of the mean WTP estimates are statistically significant at conventional significance levels. Statistically significant mean WTP coefficients include those on status quo, *homes2*, *homes3*, *wetlands* and *hard*. All coefficients on WTP standard deviations are large and statistically significant at $p \leq 0.01$, providing evidence of preference heterogeneity. This is also consistent with findings from focus groups.

Our primary research question relates to the last column of Table 2, which presents shifts in the mean coefficient estimates for the S2 sample, who were provided with percentage probabilities on inherent storm risk (compared to the S1 sample). None of these terms, included to identify the associated treatment effects, is statistically significant, suggesting no significant differences in WTP estimates between the S1 and S2 samples. This is a key finding—results of the mixed logit model suggest that the use of percentage probabilities to communicate inherent OU (compared to short-term storm frequencies alone) did not affect mean marginal WTP estimates, on average.

The lack of significant difference between results for the S1 and S2 samples implies that the values displayed in column 'Mean WTP' can be interpreted as average estimates for both samples. Since the model is estimated in WTP space, these coefficient estimates can be interpreted as annual per household WTP. Because continuous non-monetary attributes were divided by 10 prior to estimation and the cost attribute was divided by 100 to ensure convergence (see *Notes* under Table 2), the coefficient estimates for these attributes are multiplied by 10 to provide unscaled marginal WTP estimates. The resulting estimates represent per household, annual WTP (US dollars per year) for a one percentage-point change in the attribute. For example, the estimate of −1.38 on *homes2* suggests that each household is willing to pay, on average, about $13.80 annually to protect an additional percent point of homes from flooding in a Category 2 storm. The result for *homes3* implies a WTP of $12.30 per year to prevent the flooding of an additional percent point of homes in a Category 3 storm. Other estimates are interpreted similarly (with the coefficients for non-continuous variables needed to be multiplied by 100 to express unscaled WTP estimates).[20]

In summary, estimates from the mixed logit model show no evidence that the additional information used to present inherent OU through numerical probabilities (S1 versus S2) led to variations in responses or welfare estimates. One possible explanation is that respondents simply ignored the information on inherent OU—whether presented as percentages or frequencies. However, it is also possible that this information influenced responses in ways that are not immediately evident from the mixed logit estimates. For example, the large-magnitude WTP standard deviations within this model might be interpreted as evidence that a unimodal, continuous distribution of preferences—as imposed by the mixed logit model—may not be sufficient to fully capture preference and response heterogeneity. At least in part, this heterogeneity could potentially be due to whether a subject received the S1 or S2 treatment. To further explore this possibility, we continue with results from the latent class model below, which considers a different potential mechanism through which the provision of percentage probabilities might influence choice behavior.

5.2. Latent class multinomial logit model

Results of a three-class latent class multinomial logit model are presented in Table 3. The three-class model was selected based on the results of preliminary models with alternative numbers of classes.[21] The three classes identify largely distinct preferences for choice attributes. Within this model, we use the class membership function to identify how welfare estimates may be affected by the survey treatment, gender, having a college or higher degree of education, being currently employed, and residing year-round in Old

[19] The two models were estimated with sample sizes 282 and 264 respondents (or completed questionnaires), respectively, which is within the recommended SP guidelines of 200–300 respondents (e.g., [92,93]). Most of these completed questionnaires provide three choice responses, leading to 805 and 757 observations per model, respectively. The resulting sample sizes are larger than the estimated (theoretical minimum) sample sizes required to estimate model parameters at a 5 % significance level, based on S-efficiency estimates (cf. [78]). However, given the complexity of our models (i.e., large number of parameters relative to number of observations), it is conceivable that results could be influenced by statistical power and should thus be interpreted accordingly. For verification of robustness of our findings, in the Appendix, we report estimation results of a multinomial logit model, which is more restrictive but includes fewer (i.e., 18) parameters. The estimation results closely correspond to those obtained from the mixed logit model, with our main conclusions being unaffected.

[20] The statistically insignificant cost coefficient estimate cannot be interpreted as a statistically insignificant estimate on the marginal utility of income (i.e., implying that the cost is irrelevant to respondents), because the estimate refers to the underlying normal distribution (not the lognormal distribution used to model the cost).

[21] A latent-class model specification with two classes resulted in a worse model fit than the three-class model, as assessed based on the value of the log-likelihood function at convergence and the AIC. A latent-class model specification with four classes did not generate plausible results that could help distinguish distinct preference classes. The three-class model already includes a class (Class 3) with only one statistically significant DCE attribute and a model with four classes included numerous insignificant parameters.

Table 2
Results of the WTP-Space mixed logit model.

Variable	Mean WTP (Std. Error)	Standard Deviation (Std. Error)	Mean WTP Shift for S2 Sample (Std. Error)
status quo	−4.83*** (1.24)	10.34*** (3.01)	0.04 (0.48)
homes2	−1.38** (0.63)	4.18*** (1.18)	0.35 (0.68)
homes3	−1.23* (0.64)	4.47*** (1.23)	−0.44 (0.73)
wetlands	−1.32* (0.74)	3.64*** (0.99)	−0.17 (0.88)
beaches	−0.24 (0.42)	3.07*** (0.83)	−0.95 (0.61)
seawalls	−0.59 (0.38)	1.17*** (0.33)	0.50 (0.39)
hard	−1.47** (0.66)	2.16*** (0.59)	0.66 (0.61)
soft	−0.56 (0.52)	3.00*** (0.87)	0.47 (0.56)
Sign-reversed *cost*	0.46 (0.53)	1.99*** (0.43)	0.33 (0.47)
Model diagnostics			
Log-likelihood at convergence	−678.50		
Log-likelihood at constants only	−883.88		
McFadden's pseudo-R^2	0.2324		
AIC/*n*	1.8422		
BIC/*n*	2.2094		
Number of observations (*n*)	805		
Number of respondents	282		
Number of parameters	63		

Notes: * means *p*-value $\leq$ 0.10, ** *p*-value $\leq$ 0.05 and *** *p*-value $\leq$ 0.01. To ensure model convergence, all continuous attribute variables are scaled prior to estimation by dividing non-monetary attribute levels by 10 and policy cost by 100. The non-monetary attribute parameters are assumed to follow normal distributions and the cost parameter is modeled as lognormal. For the cost coefficient, the estimates for the underlying normal distribution are provided. As per equation (3), we use a negative of the cost in the model.

Table 3
Results of the WTP-Space latent class multinomial logit model.

Variable	Class 1	Class 2	Class 3
	Mean WTP (Std. Error)	Mean WTP (Std. Error)	Mean WTP (Std. Error)
status quo	0.76 (0.49)	−2.51*** (0.53)	1.58 (1.23)
homes2	1.37*** (0.50)	−0.54** (0.21)	−0.53 (0.55)
homes3	0.94** (0.44)	−0.55** (0.23)	−0.46 (0.48)
wetlands	0.11 (0.32)	−0.65** (0.26)	−0.74 (0.74)
beaches	1.16*** (0.32)	−0.22 (0.15)	−0.16 (0.39)
seawalls	0.47 (0.33)	−0.31** (0.14)	0.23 (0.32)
hard	−0.37 (0.34)	−0.46** (0.22)	−0.08 (0.52)
soft	0.62** (0.28)	−0.10 (0.20)	0.68 (0.64)
Sign-reversed *cost*	−5.42 (3.40)	1.13*** (0.29)	1.55* (0.83)
Class membership probability function			
constant	−0.61 (1.55)	1.50 (1.16)	0.00 (fixed)
S2 sample	0.60 (0.51)	0.01 (0.32)	0.00 (fixed)
female	0.63 (0.62)	1.21*** (0.34)	0.00 (fixed)
college degree or higher	−0.07 (0.50)	0.02 (0.34)	0.00 (fixed)
currently employed	0.33 (0.51)	−0.42 (0.33)	0.00 (fixed)
year-round resident of Old Saybrook	−0.80 (1.51)	−1.10 (1.13)	0.00 (fixed)
Average class probabilities			
	14.75	56.11	29.14
Model diagnostics			
Log-likelihood at convergence		−640.22	
Log-likelihood at constants only		−830.57	
McFadden's pseudo-R^2		0.2292	
AIC/*n*		1.7945	
BIC/*n*		2.0330	
Number of observations (*n*)		757	
Number of respondents		264	
Number of parameters		39	

Notes: * means *p*-value $\leq$ 0.10, ** *p*-value $\leq$ 0.05 and *** *p*-value $\leq$ 0.01. To ensure model convergence, all continuous attribute variables are scaled prior to estimation by dividing non-monetary attribute levels by 10 and policy cost by 100. As per equation (6), we use a negative of the cost in the model.

Saybrook.[22] All variables in the class membership function are defined as zero-one-coded binary indicators for the respective characteristics. Respondents' degree of education is intended as a proxy for their numeracy skills, and was allowed to influence class membership based on prior findings from the risk communication literature that numeracy can affect risk perceptions [34,35].[23] The sample size used in the model is by 18 respondents smaller than the sample used in the mixed logit model because of missing observations for some of the sociodemographic characteristics included in the latent class model. We discuss each class in a descending order of estimated class probability. Parameters in the class membership function are estimated relative to those for Class 3, which are set to zero by the model.

Class 2, which constitutes the largest share in the pooled sample (56 % probability), reveals results that are qualitatively similar to those from the mixed logit model above. Coefficient estimates for this class imply positive and statistically significant WTP for protection of wetland habitats and homes vulnerable to flooding in Category 2 and 3 storms. A household belonging to Class 2, for example, is willing to pay annually, on average, $5.40 and $5.50 to protect an additional percent point of homes from flooding in Category 2 and 3 storms, among multiple other statistically significant welfare estimates. The S2 survey treatment has no statistically significant impact on membership in Class 2 relative to Class 3 (the default).

Class 3 has an average class probability of roughly 29 %. Within this class, only the cost attribute is statistically significant. This result implies that households in Class 3 do not hold significant WTP for any type of climate change adaptation action or the protection of homes and natural habitats—their choices are driven only by program cost. This reveals a type of behavior consistent with neoclassical assumptions and common in latent class models, wherein one class is motivated primarily by cost and holds little value for environmental outcomes (e.g., [95,96]). Because Class 3 is the "default" class for the model, the statistical significance of survey treatment (S1 versus S2) on Class 3 membership is not estimated, but implied by the corresponding coefficient estimates for Classes 1 and 2.

Class 1 is the smallest predicted latent class (roughly 15 % inclusion probability), and includes several WTP parameters with signs opposite to expectations based on standard neoclassical assumptions. For example, the results suggest that households in this class would be willing to pay positive amounts for losses in the attributes *homes2*, *homes3* and *beaches*. This class is also characterized by seemingly high randomness of choices, as implied by the (relatively) large point-estimate magnitude, unanticipated negative sign, and statistically insignificant coefficient on *cost*. Because the marginal utility of income and scale are confounded in the *cost* parameter within WTP-space models, variations in choice randomness (and hence scale) are revealed through this parameter. The cause of the apparent choice randomness and unexpected welfare estimates within Class 1 is unknown, but could potentially be due to factors such as confusion, choice uncertainty, scenario rejection or protesting, the use of unanticipated decision heuristics, or other factors (cf. [51,97]). The statistically insignificant coefficient estimates on the sociodemographic variables and S2 sample again suggest that these have no effect on the probability of membership in Class 1.

In conclusion, neither the mixed logit nor the latent class model provide evidence that the inclusion of percentage probabilities in the DCE questionnaire affected WTP estimates or choice behavior. Coefficient estimates from the S1 and S2 samples are statistically indistinguishable in all cases. Similar conclusions emerge from alternative model specifications, suggesting the robustness of the presented results. These results imply that the supplemental provision of percentage probabilities did not affect respondents' preferences or values for coastal climate change adaptation policies, compared to alternative forms of risk communication based on historical event frequencies alone.

5.3. Response propensity

A final set of statistical tests is used to explore the possibility that the provision of percentage probabilities may have influenced respondents' propensity to complete the survey. To consider this possibility, we test for sociodemographic differences between the S1 and S2 survey treatments. The results of these tests are shown in Table 4. This analysis is based on the sample of 282 respondents, each of which answered at least one choice task. The last column shows that the two survey treatments are statistically significantly different only in terms of respondents' age (p-value = 0.043) and current employment status (p-value = 0.080) (two variables likely to be correlated, i.e., older respondents are more likely to have retired). Specifically, the average respondent that took the S1 survey (which omitted percentage probabilities) is somewhat older and less likely to be currently employed, relative to the average respondent that took the S2 survey (which included percentage probabilities in addition to storm frequencies).

The reason for this apparent difference is unclear, as the sample frames for both survey treatments were identical. It is possible—though seemingly unlikely—that the additional risk information presented in percentage terms could have caused a difference in the propensity to respond (to the survey) across different demographic groups.[24] This variation may also be due, however, to simple statistical chance (Type I error). Although we lack the data to explore this issue further in the present case, this intriguing finding highlights the potential relevance of future research into whether and how the approaches used to communicate OU might influence some residents' propensity to participate in a SP questionnaire.

[22] Although the results of a preliminary (three-class) latent class model that omits the sociodemographic variables suggest that the survey treatment affects the class membership, the results presented here show that this effect disappears when we control for sociodemographic characteristics. A log-likelihood ratio test also supports the latent class model with sociodemographic variables.

[23] Peters et al. [94], for example, find that people with lower numeracy skills perceive a percentage as less risky than the corresponding frequency [35].

[24] In the context of the current research, for example, it is conceivable that older, retired adults that disagreed with the shown probabilities of future storm events (S2 survey) were more reluctant to engage in this policy-relevant research study related to climate adaptation – lest this leads to higher taxes, unwelcome changes to property rights, friction between neighbors and friends, or other outcomes that increase financial and social uncertainty as they age.

Table 4
Selected sociodemographic descriptive statistics for S1 and S2 survey samples (n = 282) with tests of equality.

Sociodemographic Variable	S1 Sample (Omitted Numerical Probabilities)	S2 Sample (Included Numerical Probabilities)	p-value
Discrete Sociodemographic Variable			
Female	45.7 %	40.0 %	0.342
Graduate Degree	31.1 %	30.1 %	0.859
4-year college degree	30.4 %	29.3 %	0.839
2-year college degree	10.8 %	13.0 %	0.577
Some college	16.9 %	13.8 %	0.487
Currently Employed	55.4 %	65.9 %	0.080
Year-Round Resident of Old Saybrook	96.7 %	96.8 %	0.941
Continuous Sociodemographic Variable			
Age	62.7	59.9	0.043
Annual Household Income (USD)	119,154	126,364	0.400
Years of Residency in Old Saybrook	21.9	22.5	0.800

Notes: p-values are for the null hypothesis of no difference between S1 and S2 samples with respect to the indicated sociodemographic variable. For the discrete sociodemographic variables, the table shows the shares of respondents within each sample, and chi-squared tests of equality of proportions are used to test for differences. For the continuous variables, mean values are reported, and the Wilcoxon signed-rank test is used to test for differences. As household income was measured on an eight-point scale representing different income categories, midpoints of the categories are used.

6. Conclusions

The environmental SP literature often reports WTP estimates derived using DCE scenarios that present exogenously varying percentage probabilities as a means to communicate OU. The internal variation in these presented probabilities within these scenarios may promote internal sensitivity to this information, even if these percentage probabilities are not fully understood or accepted by all respondents. External tests are rarely conducted to evaluate whether respondents behave in ways that are consistent with an accurate understanding of this information (or even attendance to this information). These concerns can be magnified in cases of inherent OU, in which internal variation in probabilities (i.e., within a DCE) is not present.

Results of the present analysis provide no evidence that the communication of inherent OU via percentage probabilities—presented in addition to information on historical event frequencies—influenced SP responses. Hence, there is no evidence that this supplemental, numerical information helped respondents make more informed choices.[25] One possible explanation for this result is that respondents receiving this supplemental information may have overlooked the provided percentage probabilities when reading the questionnaire. This explanation, while possible, is not supported by survey pretests. Qualitative feedback from focus group participants during pretests suggested that residents paid attention to survey text that provided information about the likelihood of future storm events—this was among the most salient features of the questionnaire. Focus group results also provided evidence that raw storm frequencies were more easily understood than analogous percentage probabilities. Given preliminary qualitative input such as this, it is perhaps not surprising that DCE results were unaffected by supplemental provision of information in the form of percentage probabilities.

Taken as a whole, results of the present study suggest that further research is required to determine whether and how respondents react to information on inherent OU within SP studies. Although some past work has shown sensitivity to information of this type when combined with visual aids [21], results of the present analysis are consistent with studies outside the SP literature suggesting that individuals may not use percentage probabilities as researchers might expect [28,38,44,45,47–49].

We emphasize that the results presented here are specific to our case study of coastal adaptation in a community familiar with coastal hazards. As such, residents may have strong prior beliefs regarding associated risks. Similar findings may or may not apply to other contexts, for example, in which respondents might have more diffuse priors. Past research on the context-dependence of risk comprehension suggests that risk communication strategies might vary in effectiveness over different contexts [48,98,99]. Moreover, some sociodemographic characteristics of our respondent samples differ between the two DCE treatments, providing at least suggestive (though far from conclusive) evidence that the use of percentage probabilities to communicate inherent OU might have influenced some individuals' propensity to complete the questionnaire. Our results and conclusions should be interpreted accordingly.

This study also focuses primarily on the effect of risk *communication* and abstracts from additional challenges that can emerge when seeking to quantify inherent OU that is applicable to future time periods and might change over time. In the present case study, we expect that historical storm frequency provides a reasonable approximation of storm probability over the time period covered by the DCE scenarios (up through the mid-2020s). In such cases, one can present equivalent inherent OU information using either historical frequencies or predicted percentage probabilities. However, over longer periods, the probability of severe storms might diverge considerably from that implied by storm frequencies over a particular historical time period. Additional work would be required to identify and address additional challenges for risk communication in DCEs wherein inherent OU might change appreciably over time periods covered by survey scenarios.

[25] Note that the use of percentages to communicate probabilities is distinct from the use of percentages to communicate changes in non-probability attributes within DCEs. Whereas the risk-communication literature argues against the use of percentage probabilities to communicate risk (see citations above), the environmental DCE literature provides evidence directly supporting the use of percentages along with supporting cardinal numbers (as done here) to communicate non-probability attributes, such as the proportion of homes subject to flooding [67,68].

CRediT authorship contribution statement

Christos Makriyannis: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Supervision, Validation, Writing – original draft, Writing – review & editing. **Robert J. Johnston:** Conceptualization, Data curation, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Writing – original draft, Writing – review & editing. **Ewa Zawojska:** Formal analysis, Software, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix. Results of the WTP-Space Multinomial Logit Model

Variable	Mean WTP (Std. Error)	Mean WTP Shift for S2 Sample (Std. Error)
status quo	−0.87*** (0.25)	−0.51 (0.75)
homes2	−0.87** (0.38)	0.31 (0.84)
homes3	−0.84** (0.40)	−0.45 (1.17)
wetlands	−0.78* (0.46)	−0.23 (1.24)
beaches	−0.35 (0.24)	−1.07 (1.01)
seawalls	−0.38 (0.26)	0.18 (0.57)
hard	−0.61 (0.41)	−1.10 (1.58)
soft	−0.23 (0.35)	−0.45 (0.98)
Sign-reversed *cost*	0.56*** (0.19)	−0.29 (0.28)
Model diagnostics		
Log-likelihood at convergence	−864.36	
Log-likelihood at constants only	−883.88	
McFadden's pseudo-R^2	0.0221	
AIC/*n*	2.1922	
BIC/*n*	2.2971	
Number of observations (*n*)	805	
Number of respondents	282	
Number of parameters	18	

Notes: * means *p*-value $\leq$ 0.10, ** *p*-value $\leq$ 0.05 and *** *p*-value $\leq$ 0.01. We use a negative of the cost in the model. For consistency with the results presented in the main text of the article, all continuous attribute variables are scaled prior to estimation by dividing non-monetary attribute levels by 10 and policy cost by 100.

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